MIL-E-1/1424A(NAVY)
17 October 1969
SUPERSEDING
MIL-E-1/1424 (NAVY)
14 November 1961

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER, TYPE 6303A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Diode, rectifier
Outline - Figure 1
Base - A4-18
Cap - See figure 1
Cathode - Bonded thoria tungsten filament

Pin Number:	1	2	3	4	Cap
Element:	nc	f	f	nc	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	epx	ib	Ib	Ieff	Pp	Alt
Unit:	Vac	kv	a	mAdc	Aac	W	ft
Maximum:							
Rectifier	11.5±2.5%	40	2.5	700	---	550	10,000
TEST CONDITION:	11.5	---	---	---	---	---	---

GENERAL:

Qualification - Required

Method	Requirement or Test	Conditions	AQL (percent defective)	Inspection level	Symbol	Limits		Unit
						Min.	Max.	
	Quality conformance inspection, part 1							
1231	Peak emission	eb=8.0kv; Ef=12.2 Vac	0.65	II	is	30	--	a
1256	Electrode current (anode)	Eb=500 Vdc	0.65	II	Ib	750	--	mAdc
1301	Filament current		0.65	II	If	14.00	16.25	Aac
1353	Operation of rectifiers (1)	epx=40 kv; Ib=700 mAdc; ib=2.5 a (see note 1)	0.65	II	--	--	--	--

FSC 5960

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Method	Requirement or Test	Conditions	AQL (percent defective)	Inspection level	Symbol	Limits		Unit
						Min.	Max.	
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	No voltage	6.5	S3	--	--	--	--
1041	Shock test	300 G	6.5	S3	--	--	--	--
--	Operation of rectifiers (2)	Ef=12.2 Vac (see notes 1 and 2)	6.5	S3	--	--	--	--
--	Slap on voltage	Note 3						
--	Crowbar test	Note 4						
	Quality conformance inspection, part 3							
--	Life test	Group D; operation of rectifiers (1)	--	--	t	500	--	hr
--	Life-test end point	Peak emission	--	--	is	20	--	a

NOTES:

1. The tube shall operate for 2 minutes without arcing or sign of gas during a test interval not to exceed 5 minutes.

2. The tube shall be operated as a shunt diode in a thyratron modulator. The circuit constants shall be chosen to give the following diode operating conditions: epx = 33kv; ib = 50 a; and effective current (Ieff) = 1.25 Aac.

3. A slap-on voltage shall be applied to the tube under the following load condition: epx = 30 kv; Ib = 700 mAdc; ib = 6.0 a minimum for a minimum duration of 16 milliseconds. The slap-on voltage shall be applied as often as necessary during a test interval not to exceed 5 minutes until the tube hold-off is five times without kicking out the overcurrent relay.

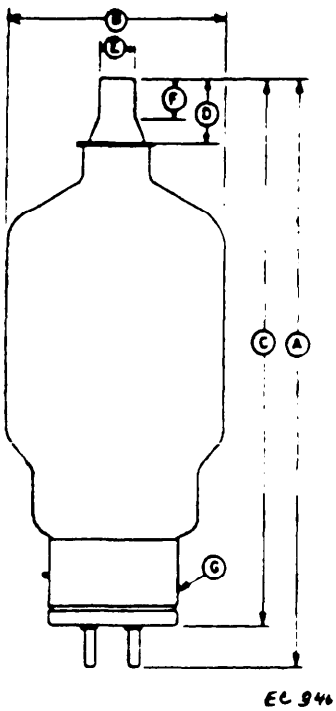
4. While operating at full load as a rectifier under the following conditions, epx = 40 kv; Ib = 700 mAdc; ib = 2.5 a, the tube shall be subjected to a simulated crowbar test by drawing a 12.5 a short 20 times. These crowbars shall not result in damage to the cathode coating. Minimum of crowbar shall be 32 milliseconds. Following this test the tube shall meet operation (1) requirements.

User activities:
Navy-AS, OS, MC, CG, SH

Preparing activity:
Navy-EC

(Project 5960-N824-8)

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DIM.	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL	LIMITS	
			Min.	Max.
QUALIFICATION INSPECTION				
D	–	–	0.969	1.031
E	–	–	0.559 dia	0.573 dia
F	–	–	0.594	0.656
G	Base: A4-18(see note b)			
QUALITY CONFORMANCE INSPECTION, PART 2				
A	6.5	S3	9.250	9.750
B	6.5	S3	–	3.625 dia
C	6.5	S3	8.472	9.032

NOTES:

a. All dimensions in inches.
b. For pin alinement use gage GA4-7.

Figure 1 - Outline drawing of tube type 6303A.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the s ification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for taining information on the use of this specification which will insure that suitable products can be procured with minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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